

# Simplifying algebraic expressions

Collecting like terms, and taking a common factor out of a bracket.

LESSON 11–12

2 × 40 MIN

MATEMATIKA 6, §2

S7 2.3

LESSON CLOCK

20'

25'

20'

12'

3

|                     |        |
|---------------------|--------|
| Like terms          | 20 min |
| Collecting          | 25 min |
| Expanding a bracket | 20 min |
| Taking a factor out | 12 min |
| Homework            | 3 min  |

LEARNING OBJECTIVES

- Identify like terms and collect them.
- Expand a bracket by multiplying every term inside it.
- Take a common factor outside a bracket.
- Check a simplification by substituting a number.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 29–34          |
| Cambridge | Stage 7, pp. 25–29 |
| Workbook  | Exercise 2.3       |

01

## Explanation

### Like terms

Terms are **like** when their letter parts are identical. Only like terms can be collected — for the same reason that apples and oranges cannot be added into one number of apples.

| PAIR            | LIKE? | WHY               |
|-----------------|-------|-------------------|
| $5x$ and $-2x$  | yes   | the same letter   |
| $5x$ and $5y$   | no    | different letters |
| $5x$ and $5x^2$ | no    | different powers  |
| $3ab$ and $7ba$ | yes   | $ab = ba$         |

THE COEFFICIENT MAY DIFFER; THE LETTERS MAY NOT

$5x$  and  $-2x$  are like terms and collect to  $3x$ .  $5x$  and  $5y$  are not, and  $5x + 5y$  is already as simple as it gets.

Collecting

| EXPRESSION         | COLLECTED |
|--------------------|-----------|
| $5x + 3x$          | $8x$      |
| $7a - 2a$          | $5a$      |
| $4x + 3y + 2x - y$ | $6x + 2y$ |
| $6m - 9m$          | $-3m$     |
| $3a + 4b$          | $3a + 4b$ |

A TERM CARRIES THE SIGN IN FRONT OF IT

In  $4x + 3y + 2x - y$  the last term is  $-y$ , with coefficient  $-1$ . Collecting the  $y$  terms gives  $3y - y = 2y$ , not  $4y$ .

Expanding a bracket

$$a(b + c) = ab + ac$$

| EXPRESSION      | EXPANDED   |
|-----------------|------------|
| $3(x + 4)$      | $3x + 12$  |
| $5(2a - 3)$     | $10a - 15$ |
| $x(x + 2)$      | $x^2 + 2x$ |
| $2(x + 3) + 4x$ | $6x + 6$   |

EVERY TERM INSIDE THE BRACKET IS MULTIPLIED

$3(x + 4)$  is  $3x + 12$ , not  $3x + 4$ . Drawing two arrows from the 3 to both terms stops this error for good.

Taking a factor out

Expanding read backwards: find what every term has in common and write it outside.

| EXPRESSION | COMMON FACTOR | FACTORISED   |
|------------|---------------|--------------|
| $6x + 9$   | 3             | $3(2x + 3)$  |
| $4a - 10$  | 2             | $2(2a - 5)$  |
| $x^2 + 5x$ | $x$           | $x(x + 5)$   |
| $12m + 8n$ | 4             | $4(3m + 2n)$ |

CHECK BY SUBSTITUTING A NUMBER

Put  $x = 2$  into  $6x + 9$  and into  $3(2x + 3)$ : both give 21. One substitution catches almost every slip in this lesson.

02

## Worked examples

EXAMPLE 1 Simplify  $4x + 3y + 2x - y$ .

①  $x$  terms:  $4x + 2x = 6x$ .

②  $y$  terms:  $3y - y = 2y$ .  
The last term is  $-y$ .

③  $= 6x + 2y$   
Nothing else is like.

ANSWER

$6x + 2y$

**EXAMPLE 2** Simplify  $2(x + 3) + 4x$ .

- ① Expand:  $2x + 6 + 4x$ .  
Both terms multiplied.
- ② Collect:  $6x + 6$ .
- ③ Check at  $x = 1$ :  $8 = 8$  ✓

ANSWER

$6x + 6$

**EXAMPLE 3** Factorise  $6x + 9$ .

- ① Both terms divide by 3.
- ②  $6x \div 3 = 2x$  and  $9 \div 3 = 3$ .
- ③  $= 3(2x + 3)$   
Check by expanding.

ANSWER

$3(2x + 3)$

**03** Interactive model

Use coloured chalk for each letter when collecting; the class sees the groups before it counts them, and the sign errors halve.

**Collect, expand, factorise**

$5x + 3x$  equals:

$8x$

$8x^2$

$15x$

$8$

$5x$  and  $5y$  collect to:

$10xy$

$10x$

they do not collect

$25xy$

$4x + 3y + 2x - y$  equals:

$6x + 4y$

$6x + 2y$

$8x + 2y$

$6x - 2y$

$6m - 9m$  equals:

$3(x + 4)$  equals:

$5(2a - 3)$  equals:

$6x + 9$  factorises to:

$x^2 + 5x$  factorises to:

$$x(x + 5)$$

$$x(x + 5x)$$

$$5x(x + 1)$$

$$x^2(1 + 5)$$

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH            | O'ZBEKCHA           | РУССКИЙ               |
|--------------------|---------------------|-----------------------|
| To simplify        | Soddalashtirish     | Упростить             |
| Like terms         | O'xshash hadlar     | Подобные члены        |
| To collect         | Ixchamlash          | Приведение подобных   |
| To expand          | Qavsni ochish       | Раскрыть скобки       |
| Common factor      | Umumiy ko'paytuvchi | Общий множитель       |
| Bracket            | Qavs                | Скобка                |
| Coefficient        | Koeffitsiyent       | Коэффициент           |
| Substitution check | Tekshirish          | Проверка подстановкой |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

Like terms have:

the same coefficient

the same letter part

the same sign

nothing in common

$7a - 2a$  equals:

$5a$

$9a$

$5$

$14a$

$3a + 4b$  simplifies to:

$7ab$

$7a$

it is already simplest

$12ab$



$x(x + 2)$  equals:

$4a - 10$  factorises to:

A simplification is checked by:





## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1.  $5x + 3x$

ANSWER  $8x$ 

2.  $7a - 2a$

ANSWER  $5a$ 

3.  $6m - 9m$

ANSWER  $-3m$ 

4.  $3(x + 4)$

ANSWER  $3x + 12$ 

5.  $5(2a - 3)$

ANSWER  $10a - 15$ 

6.  $6x + 9$  factorised

ANSWER  $3(2x + 3)$ 

7.  $3a + 4b$

ANSWER Already simplest

Medium · the standard question

7 PROBLEMS

1.  $4x + 3y + 2x - y$

ANSWER  $6x + 2y$

2.  $2(x + 3) + 4x$

ANSWER  $6x + 6$

3.  $x(x + 2)$

ANSWER  $x^2 + 2x$

4.  $x^2 + 5x$  factorised

ANSWER  $x(x + 5)$

5.  $12m + 8n$  factorised

ANSWER  $4(3m + 2n)$

6.  $5a + 2b - 3a + 4b$

ANSWER  $2a + 6b$

7.  $3(2x - 1) + 5$

ANSWER  $6x + 2$

### Hard · stretch and reason

7 PROBLEMS

1.  $4(x + 2) - 3(x - 1)$

ANSWER  $x + 11$ 

2.  $2x(x + 3) - x^2$

ANSWER  $x^2 + 6x$ 

3.  $5(a + b) - 5a$

ANSWER  $5b$ 

4.  $3x + 2y - (x - y)$

ANSWER  $2x + 3y$ 

5. Factorise  $9x^2 + 6x$

ANSWER  $3x(3x + 2)$ 

6. Check  $6x + 9 = 3(2x + 3)$  at  $x = 2$

ANSWER Both give 21

7. Why can  $3a$  and  $3a^2$  not be collected?

ANSWER Their letter parts differ

**07 Homework****Homework — 5 tasks**

Substitute one number into both forms at the end; if they disagree, the simplification is wrong.

1. Simplify  $7x + 2y - 3x + 5y$ .
2. Expand  $4(x + 5)$  and  $6(2a - 3)$ .
3. Simplify  $3(x + 2) + 2x$ .
4. Factorise  $8x + 12$  and  $x^2 + 7x$ .
5. Check your answer to task 4 by substituting  $x = 3$  into both forms.